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SUPERSEDING
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FEDERAL SPECIFICATION

PIPE FITTINGS, FLANGE FITTINGS, AND FLANGES: STEEL AND MALLEABLE IRON (THREADED AND BUTT-WELDING) CLASS 150

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fittings, threaded and butt welding, flanged fittings, and flanges, class 150.

1.2 Classification. The fittings shall be of the following types as specified (see 6.2).

Type I - Black (uncoated natural finish).
Type II - Zinc-coated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

FSC 4730

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-V-3 - Valve, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications, and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS):

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI):

B2.1 - Pipe Threads (Except Dryseal).
B16.3 - Malleable-Iron Threaded Fittings.
B16.5 - Steel Pipe Flanges, Flanged Valves, and Fittings.
B16.9 - Factory-Made Wrought Steel Buttwelding Fittings.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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American Society for Testing and Materials (ASTM) Standards:

A153 - Zinc-Coating (Hot-Dip) on Iron and Steel Hardware.
A164 - Electrodeposited Coatings of Zinc on Steel.
A181 - Forged or Rolled Steel Pipe Flanges, Forged Fittings, and Valves and Parts for General Service.
A197 - Cupola Malleable Iron.
A216 - Carbon-Steel Castings Suitable for Fusion Welding for High Temperature Service.
A234 - Pipe Fittings for Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. All material shall be as specified herein. Fittings, flanged fittings, and flanges shall be made of commercial grade malleable iron, or carbon steel (see 6.2). Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

3.1.1 Malleable iron fittings. Malleable iron fittings shall be made from malleable iron conforming to ASTM A197.

3.1.2 Steel fittings. Steel fittings shall be made from steel conforming to ASTM A216 or A234.

3.1.3 Steel flanges. Flanges shall be made from material conforming to ASTM A181.

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3.2 Standard commercial product. The pipe fittings, flanged fittings, and flanges shall, as a minimum, be in accordance with the requirements of this specification and may be the manufacturer's standard commercial product, new and unused, with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product may be included in the items being furnished. Standard commercial product is a product which has been sold on the commerical market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.3 Performance characteristics. Primary service rating shall be 150 pounds per square inch (psi) maximum saturated steam pressure.

3.3.1 Threaded fittings. Threaded fittings shall conform to ANSI B16.3, class 150.

3.3.2 Welded fittings. Welded fittings shall conform to ANSI B16.9.

3.3.3 Flanges and flange fittings. Flanges and flange fittings shall conform to ANSI B16.5.

3.4 Detail requirements. The kind of fitting shall be as specified (see 6.2), and shall correspond to ANSI B16.3, B16.5, or B16.9 as applicable.

3.4.1 Size. The size of the fittings shall be specified in terms of nominal pipe size (see 6.2).

3.4.2 Threads. Threaded fittings shall conform to NBS H28 or ANSI B2.1 American Standard Taper Pipe Threads, except that couplings and caps in nominal sizes 3/8 and smaller, when made of steel bar stock, may have straight pipe threads, American Standard Straight Pipe Coupling. Threads shall be righthand unless otherwise specified (see 6.2).

3.4.2.1 Threaded ends. Each internally threaded end of malleable iron fittings shall have a band, rectangular in section, with or without rounded corners, and with the face either flat or beveled not more than 5°.

3.5 Type II fittings. Zinc coating shall be either by the hot-dip or electroplating process.

3.5.1 Hot-dip process. Coating by the hot-dip process shall conform to ASTM A153. Fittings shall be coated before machining and threading.

3.5.2 Electroplating process. Coating by the electroplating process shall conform to QQ-Z-325, type I, class 2; or ASTM A164, type LS. Fittings shall be coated after machining and threading.

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3.6 Identification marking. The manufacturer's name, initials, or trademark, shall be legibly cast or die-stamped on each fitting.

3.7 Workmanship.

3.7.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.7.2 Zinc coating. The fittings shall be thoroughly coated with a smooth, bright, adherent, solid surface of zinc, free from blisters, dross, dirt, bare spots, rust, and other defects in the coating. Fittings shall be zinc-coated inside and outside.

3.7.3 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. For the purpose of sampling, a lot shall consist of all fittings of the same type, kind, and size offered for delivery at one time.

4.2.1 Sampling for examination. A random sample shall be selected from each lot in accordance with MIL-STD-105. The inspection shall be level II with an Acceptable Quality Level (AQL) of 1.5 percent defective.

4.2.2 Sampling for test. When specified (see 6.2), a random sample shall be selected from each lot in accordance with MIL-STD-105. The inspection shall be level S-4 with an AQL of 1.0 percent defective.

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4.3 Examination. Each sample selected in accordance with 4.2.1 shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass visual examinations and dimensional measurements.

4.4 Tests. Each sample selected in accordance with 4.2.2 shall be tested for compliance to this specification by one of the following pressure tests.

4.4.1 Air pressure. The fittings shall be subjected for 10 seconds, while under water or other suitable fluid, to an air pressure of at least 60 psi gage. Any leakage of air shall be cause for rejection.

4.4.2 Hydrostatic pressure. The fitting shall be subjected for 10 seconds to a hydrostatic pressure of 300 psi gage. Any leakage of water or sweat at any part of the surface shall be cause for rejection.

4.5 Inspection of preparation for delivery. The preservation, packaging, and marking of the pipe fittings, flange fittings, and flanges shall be inspected to verify conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The pipe fittings and flanges shall be preserved and packaged in accordance with the applicable level A requirements of MIL-V-3.

5.1.2 Level C. The pipe fittings and flanges shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The pipe fittings and flanges shall be packed in accordance with the level A requirements of MIL-V-3.

5.2.2 Level B. The pipe fittings and flanges shall be packed in accordance with the level B requirements of MIL-V-3.

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5.2.3 Level C. The pipe fittings and flanges shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Steam service. Pipe fittings, flanges fittings, and flanges are intended for steam service when the working pressure does not exceed 150 psi and the temperature does not exceed 350° Fahrenheit.

6.1.2 Other service. Pipe fittings, flange fittings, and flanges are also intended for water, air, gas, oil and other fluids where the pressure does not exceed 300 psi nonshock, and the temperature does not exceed 150°F.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents.

(a) Title, number, and date of this specification.
(b) Type of fittings required (see 1.2).
(c) Material required (see 3.1).
(d) Kind of fitting required (see 3.4).
(e) Size of fitting required (see 3.4.1).
(f) When left-hand threads are required (see 3.4.2).
(g) When pressure test is required (see 4.2.2).
(h) Selection of applicable level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(i) Marking required (see 5.3).

6.3 This specification contemplates the purchase of fittings by the piece. It is recommended that bids be requested on that basis.

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6.4 When reducing fittings are ordered, the kinds of fittings and the sizes of connecting ends should be specified. When reducing tees are specified, the larger run is mentioned first, the smaller run next, and the outlet or branch last.

6.5 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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